
Notes on the Biology of a Spiny Lobster, *Panulirus Guttatus*, in Bermuda

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Source: *Ecology*, Vol. 34, No. 4 (Oct., 1953), pp. 794-796

Published by: Wiley on behalf of the Ecological Society

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Accessed: 16-04-2026 17:12 UTC

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NOTES AND COMMENTS

NOTES ON THE BIOLOGY OF A SPINY LOBSTER, *PANULIRUS GUTTATUS*, IN BERMUDA¹

Various records were kept on *Panulirus guttatus* during the course of an investigation of the biology of the common Bermuda spiny lobster, *P. argus*. Other than locality records in faunistic or taxonomic treatments, not much is known of the biology of this interesting little palinurid. For this reason, it is hoped that the following observations, covering the period of October, 1951, through October, 1952, may be of value. *Panulirus guttatus* was first reported from Bermuda by Verrill in 1922.

The animals were taken in lobster traps of local design (see Sutcliffe 1952) at various localities about the Bermuda plateau (Fig. 1), but most records were obtained from a trapping section including North Rock, Castle Harbor, and Nonsuch Island. Traps were maintained at depths up to 20 fathoms. All measurements are in terms of carapace length, in mm.

Panulirus guttatus is known locally as the

¹ Contribution No. 203 from the Bermuda Biological Station.

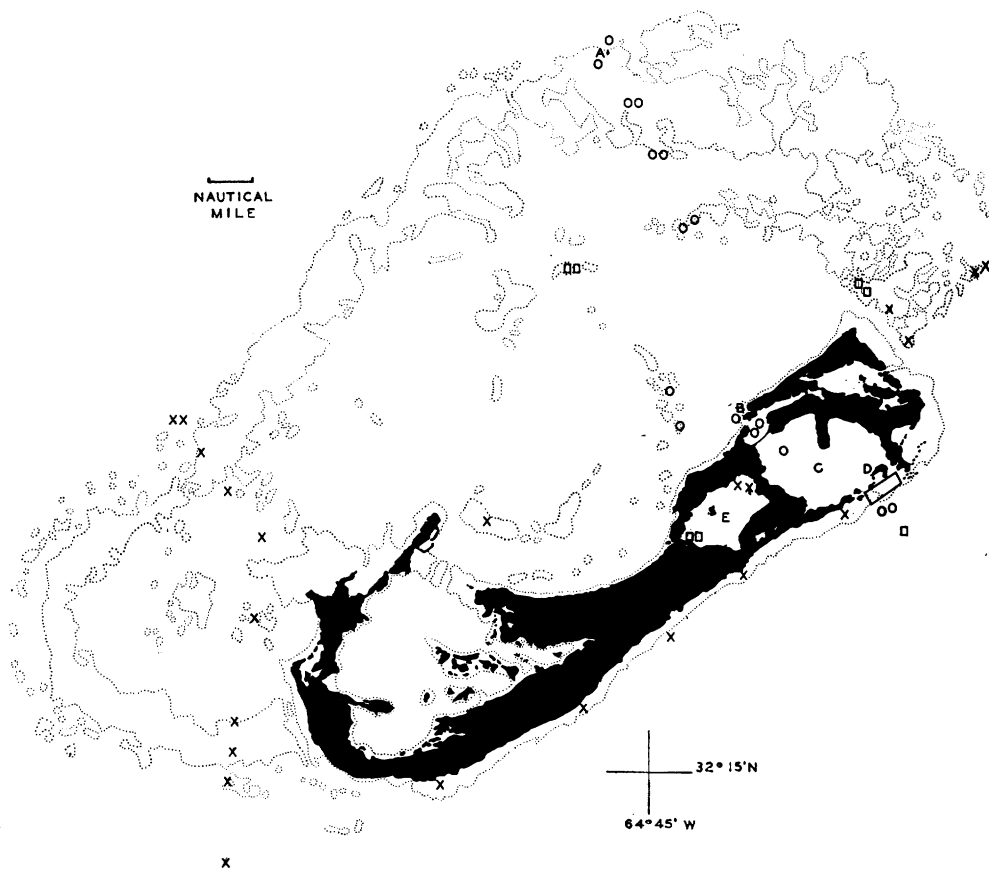


FIG. 1. Sketch map of Bermuda and reefs showing locations of traps:

Enclosed area represents seven or more traps maintained year round; isolated year round traps (O); maintained summer 1952 (x); maintained intermittently (□); North Rock (A); Whalebone Bay (B); Castle Harbor (C); Nonsuch Island (D); Harrington Sound (E). The dotted lines represent the three-fathom contour.

"guinea chick lobster" or "star lobster." Although nowhere really plentiful in Bermuda, it is sufficiently common to find a ready sale on the market, and is fairly frequently taken in traps located on the outer reefs. There it is found in a ratio of about one in twenty to *P. argus*. Unlike *P. argus* it does not seem to undergo any noticeable migratory behavior (Sutcliffe 1952), for it is seldom taken outside of a narrow band encompassing the outermost reefs or the relatively shallow water immediately adjacent. This agrees with the observations of local fishermen. Of 158 specimens examined during the period of observation, only two were found inside this reef area: one in Whalebone Bay and one in Harrington Sound. The latter specimen was caught behind the Government Aquarium and, according to Mr. Louis S. Mowbray, the Director, it was undoubtedly liberated there from the Aquarium a short time before. Adequate information is lacking regarding waters deeper than 20 fathoms outside the reefs, but in view of the very low catches per unit effort in 10 to 20 fathoms outside the reefs, as compared to the reefs proper, one wonders if occurrences in water deeper than 20 fathoms would be numerically significant.

Panulirus guttatus in Bermuda is a smaller species than *P. argus*. It may be seen (Fig. 2) that males much larger than 86 mm are not found, while in *P. argus* males attain a size well over 200 mm. If the size in the sample obtained may be considered indicative, the females are generally smaller, as with *P. argus*.

Of interest is the apparent considerable disparity in the numbers of the sexes. As seen in Figure 2, the ratio of males to females is about ten to one. This difference probably is not due to differential trapping effects, for the size of trap mesh used would retain a lobster somewhat smaller than 50 mm, while all of the captured females were 58 mm or larger. The few females taken were trapped well within the reef

area already mentioned and were not restricted to any one season (see Table I). Although there is no evidence to indicate geographical or seasonal segregation, it is possible, nevertheless, that there is a localization of females somewhere on the Bermuda plateau. An interesting consideration is the possibility of differential survival of the two sexes inasmuch as Bermuda represents the northernmost known limit of this species.

TABLE I. Numbers of females of *Panulirus guttatus* with seasonal distribution and breeding condition.

Date	Bearing eggs	Having shed eggs	No signs of breeding
Oct. 1951	..	1	2
Jan. 1952	..	..	1
Feb. 1952	..	..	4
Apr. 1952	..	..	1
May 1952	..	..	1
Aug. 1952	1	1	
Sept. 1952	..	2	1

Table I gives the available information concerning breeding. The one female with eggs was found in August, 1952; four females bearing a spermatophore remnant and showing signs of having carried eggs were taken in August, 1952, September, 1952, and October, 1951.

SUMMARY

1. The distribution of *P. guttatus* in Bermuda is relatively sparse and is apparently limited to the outermost reefs. This lobster is almost completely lacking in the lagoon and inside waters.

2. There is as yet no evidence of migratory behavior such as found in *P. argus*, the common spiny lobster in Bermuda.

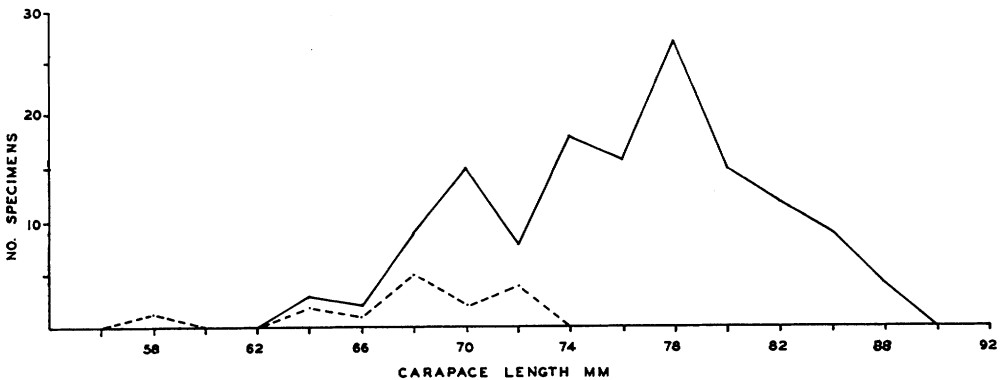


FIG. 2. Length-frequencies of *Panulirus guttatus* taken in Bermuda: females (---), males (—).

3. Present data show a disproportionate sex ratio of about ten to one of males to females, and females are generally of smaller size.

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ARE OUR STEPPES BECOMING DRIER?

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[*Editorial note.* In reading this important paper by the late L. S. Berg, the American ecologist may be assisted by a summary of European postglacial stratigraphy. During and just after the maximum advance of the ice sheets of the last (Würm) glacial age, extensive deposits of loess were laid down in the drier and warmer parts of Europe and Siberia. This loess is hardly represented in Scandinavia and northwestern Europe, where the classical pollen stratigraphy has been worked out. Conversely, in the steppe regions where the loess is mainly found there is relatively little information about later events in the form of pollen sequences from bogs and lakes. The problem discussed by Academician Berg is therefore not a straightforward one of referring various sorts of evidence (pedologic, geomorphic, paleolimnologic, biogeographic) of arid or humid climates to the appropriate level of a standard pollen diagram; such a diagram does not exist for the region in question.

So far as the climatic history of northwestern Europe is relevant (and there are good reasons for believing it to be capable of world-wide application in its broadest outline), it may be considered tripartite (von Post, L. 1946. *New Phy-*

tologist, 45: 193-217). The postglacial climate was (1) increasingly warm, (2) reached maximum warmth, and (3) "deteriorated" or became cooler. Within (2), the postglacial thermal maximum, conditions were (2a) dry, Boreal; (2b) moist, Atlantic; and (2c) dry, Sub-Boreal. Phase (3), the Sub-Atlantic, was mainly moist as well as cool. The "xerothermic" time assumed by Berg may have been either (2a) or (2c), but was probably the latter, prevailing roughly from 2000 B.C. to 500 B.C.

A more refined stratigraphy of Swedish raised bogs (Granlund, E. 1932. *Sver. Geol. Unders., Årsbok* 26 (1): 1-193) shows at least five cyclic recurrences of dry and moist climates, each resulting in swamping of a previously dry and oxidized bog surface. The swamping times are dated on archaeological evidence at 2500 B.C., 1200 B.C., 600 B.C., 400 A.D., and 1200 A.D., with the transition from Sub-Boreal to Sub-Atlantic ((2c) to (3) in the preceding paragraph) being the middle one (600 B.C., number III in Granlund's series of recurrence surfaces, which unfortunately is numbered from the youngest downward).

It follows that not all the evidence of former aridity discussed by Berg need necessarily be assigned to the Sub-Boreal ((2c), or the time just preceding Granlund's III). The deposition of the loess was almost certainly older, as Berg maintains, but Berg does not discuss the pos-

¹ Translated in slightly abridged form from the Russian by J. S. Joffe, Rutgers University, College of Agriculture. The paper appeared in *Pochvovedenie* (U.S.S.R.), No. 10, pp. 636-642 (1950).